

Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

Patient Name:	Radhika Suresh	Report ID:	APL-MULT-47-RAD-2026
Age / Gender:	47 Yrs / Female	Category:	Multi-condition: PCOS + Iron Deficiency
Weight / Height:	74 kg / 162 cm	BMI:	28.2 kg/m2
Ref. Doctor:	Dr. Arjun Mehta	Report Date:	18-Jan-2026

1. COMPLETE BLOOD COUNT [Circulatory System]

Red Cell Indices

Test	Result	Unit	Reference Range	Flag
Haemoglobin (Hb)	8.2	g/dL	12.0 - 16.0	LOW
Packed Cell Volume (PCV)	31.8	%	36 - 46	LOW
MCV	74.6	fL	83 - 101	LOW
MCH	24.2	pg	27 - 32	LOW
RDW-CV	16.5	%	11.6 - 14.0	HIGH

Haematology Comment: Haemoglobin 8.2 g/dL - microcytic (MCV 74.6 fL) hypochromic (MCH 24.2 pg) anaemia pattern. Raised RDW-CV (16.5%) indicates anisocytosis. Consistent with iron deficiency or thalassaemia trait. Further iron studies recommended.

White Cells & Platelets

Test	Result	Unit	Reference Range	Flag
Total WBC	7000	cells/cmm	4500 - 11000	Normal
Neutrophils	62	%	40 - 75	Normal
Lymphocytes	28	%	20 - 45	Normal
Platelet Count	2.8	lakh/cmm	1.5 - 4.0	Normal

Comment: WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

2. IRON STUDIES [Haematopoietic System]

Test	Result	Unit	Reference Range	Flag
Serum Iron	51	ug/dL	60 - 170	LOW
TIBC	409	ug/dL	250 - 370	HIGH
Transferrin Saturation	10.6	%	20 - 50	LOW
Serum Ferritin	10.0	ng/mL	12 - 150	LOW

Iron Studies - Critical Findings: Serum iron 51 ug/dL and ferritin 10 ng/mL are reduced. Transferrin saturation 10.6% (normal >20%).
RECOMMENDATION: Iron supplementation 60 mg elemental iron/day. Investigate dietary adequacy and occult GI blood loss.

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3. HEPATIC FUNCTION TEST [Hepatobiliary System]

Liver Enzymes

Test	Result	Unit	Reference Range	Flag
SGPT / ALT	22	U/L	7 - 56	Normal
SGOT / AST	20	U/L	10 - 40	Normal
Alkaline Phosphatase (ALP)	70	U/L	44 - 147	Normal
Total Bilirubin	0.6	mg/dL	0.2 - 1.2	Normal
Albumin	4.2	g/dL	3.5 - 5.0	Normal

Hepatologist Note: Liver enzymes within normal limits. No hepatocellular damage or cholestatic process detected.

4. RENAL FUNCTION TEST [Urinary System]

Test	Result	Unit	Reference Range	Flag
Serum Creatinine	0.9	mg/dL	0.6 - 1.1	Normal
Blood Urea Nitrogen	14	mg/dL	7 - 20	Normal
eGFR (CKD-EPI)	98	mL/min/1.73m2	60 - 120	Normal
Serum Uric Acid	4.8	mg/dL	2.4 - 6.0	Normal
Sodium (Na+)	139	mEq/L	135 - 145	Normal
Potassium (K+)	4.0	mEq/L	3.5 - 5.1	Normal

Nephrologist Note: eGFR 98 mL/min/1.73m2 - glomerular filtration well preserved. Renal function normal.

5. LIPID PROFILE [Cardiovascular Risk]

Test	Result	Unit	Reference Range	Flag
Total Cholesterol	229	mg/dL	0 - 200	HIGH
HDL Cholesterol	41	mg/dL	50 - 999	LOW
LDL Cholesterol	154	mg/dL	0 - 100	HIGH
Triglycerides	187	mg/dL	0 - 150	HIGH

Cardiologist Note: LDL 154 mg/dL (target <100 mg/dL) and Total Cholesterol 229 mg/dL are elevated. Triglycerides 187 mg/dL. Increased cardiovascular risk. RECOMMENDATION: Mediterranean diet, aerobic exercise, statin therapy consideration if >10-year CV risk >7.5%.

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6. THYROID FUNCTION TEST [Endocrine System]

Test	Result	Unit	Reference Range	Flag
TSH	4.3	uIU/mL	0.4 - 4.0	HIGH
Free T3	3.2	pg/mL	2.3 - 4.2	Normal
Free T4	1.1	ng/dL	0.8 - 1.8	Normal

Endocrinologist Comment: TSH: 4.3 uIU/mL is elevated - consistent with hypothyroidism. FT4 1.1 ng/dL is borderline (subclinical hypothyroidism). Thyroid antibodies (Anti-TPO) and ultrasound recommended. Consider levothyroxine therapy.

7. DIABETES SCREENING [Endocrine - Pancreas]

Test	Result	Unit	Reference Range	Flag
Fasting Blood Glucose	110	mg/dL	70 - 100	HIGH
Post-Prandial Glucose (2hr)	142	mg/dL	0 - 140	HIGH
HbA1c	5.7	%	0 - 5.7	Normal
HOMA-IR	3.7		0 - 2.5	HIGH

Endocrinology Note: Glucose metabolism normal. HbA1c 5.7% - no diabetes or pre-diabetes. HOMA-IR 3.7 - no insulin resistance.

8. VITAMINS & BONE [Nutritional / Skeletal]

Test	Result	Unit	Reference Range	Flag
Vitamin D (25-OH)	17.3	ng/mL	30 - 100	LOW
Vitamin B12	450	pg/mL	211 - 911	Normal
Folate (Serum)	12.0	ng/mL	5.4 - 999	Normal
Calcium (Total)	9.4	mg/dL	8.5 - 10.5	Normal
PTH	32	pg/mL	15 - 65	Normal

Nutritionist/Endocrinologist Comment: Vitamin D: 17 ng/mL - deficient (10-20). PTH at 32 pg/mL is within range. Supplementation: 60,000 IU cholecalciferol weekly x 8 weeks, then 1000-2000 IU/day. Recheck in 3 months.

9. INFLAMMATORY MARKERS [Immune System]

Test	Result	Unit	Reference Range	Flag
ESR	21	mm/hr	0 - 20	HIGH
CRP	2.4	mg/L	0 - 5	Normal
hs-CRP	1.3	mg/L	0 - 1.0	HIGH

Immunology Comment: CRP 2.4 mg/L and ESR 21 mm/hr are elevated. hs-CRP 1.3 mg/L - intermediate CV risk (1-3). Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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CLINICAL SUMMARY & RECOMMENDATIONS

[ABNORMAL]	Anaemia: Hb=8.2 g/dL (LOW). Iron deficiency pattern.
[CRITICAL]	Iron Deficiency: Serum Iron=51 ug/dL, Ferritin=10 ng/mL. Begin iron supplementation.
[ELEVATED]	LDL=154 mg/dL (HIGH). Total Cholesterol=229. Dyslipidaemia - dietary + pharmacological management.
[ABNORMAL]	Hypothyroidism: TSH=4.3 uIU/mL. Levothyroxine therapy evaluation required.
[DEFICIENT]	Vitamin D=17 ng/mL (LOW). Supplementation required.
[ACTION]	Next appointment: Dr. Arjun Mehta in 4-8 weeks for follow-up and review of supplementation response.

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Arjun Mehta

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